Regularized Multivariate Two-way Functional Principal Component Analysis

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BE THE DIFFERENCE.

Outline

1. Introduction & Background
2. A SVD Approach for Regularized Multivariate FPCA
3. Two-way Regularized Multivariate FPCA
4. Conclusion & Future Work

Introduction

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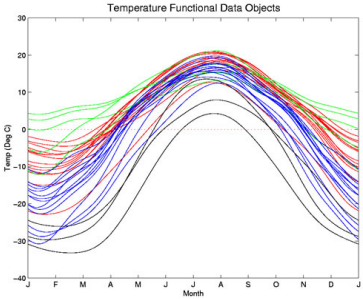

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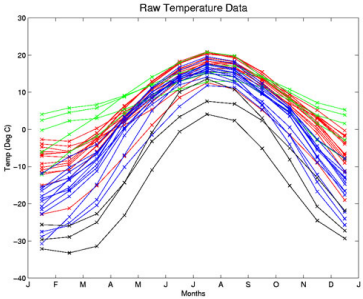

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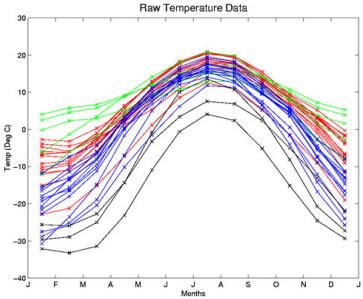
- **Functional Data Analysis (FDA)** models these observations as smooth underlying functions, improving interpretation and analysis of continuous variation.



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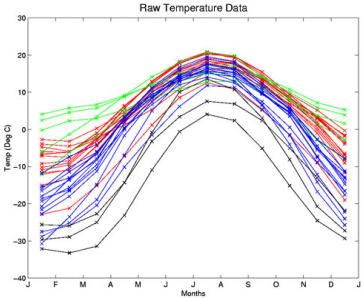
- **Functional Data Analysis (FDA)** models these observations as smooth underlying functions, improving interpretation and analysis of continuous variation.
- **Functional PCA (FPCA)** extends classical PCA to functional data by identifying orthogonal functional components that capture the main modes of variation for dimension reduction [@ramsay2005functional].



Functional Principal Component Analysis

- **Extensions of FPCA:**
  - Smoothed FPCA: Adds roughness penalties [Silverman, 1996, Huang et al., 2008].
  - Sparse FPCA: Enforces sparsity for interpretability [Shen and Huang, 2008, Nie and Cao, 2020].
  - Multivariate FPCA (MFPCA) [Silverman, 1996, Happ and Greven, 2018].
  - Regularized MFPCA: Penalties improve estimation & interpretability [Haghbin et al., 2025].
  - **Two-way functional data:** Two-way functional data consist of a data matrix whose row and column domains are both structured. Observations vary along these two domains (e.g., time $\times$ frequency), with applications in climate science, neuroscience, etc. Huang [Jianhua Z. Huang and Buja, 2009] applied regularization to both left and right singular vectors in SVD.
- **Impact:** More adaptable, robust, and applicable across different problems.

Foundations of FPCA through Minimizing Reconstruction Error

- **Goal:** Capture the dominant functional variation in $X \in \mathbb{R}^{n \times m}$ using a low-rank representation, where n is the number of discretized functions and m is the number of grid points. The vector $v \in \mathbb{R}^m$ represents the principal component function, while $u \in \mathbb{R}^n$ gives the corresponding scores.
- **Reconstruction problem:**
$$\min_{u,v} \|X - uv^\top\|_F^2 = \min_{u,v} \operatorname{tr}\{(X - uv^\top)(X - uv^\top)^\top\}.$$
- **Extensions of FPCA via Regularization:**
$$\min_{u,v} \|X - uv^\top\|_F^2 + \mathcal{P}(u, v)$$
  - **Smooth FPCA:** adds roughness penalty on functions.
  - **Sparse FPCA:** adds sparsity penalty on loadings.
- **Algorithms:** Based on iterative **power method** and **thresholding** updates.

Smooth Functional PCA

- **Problem setup** [Huang et al., 2008]: Smooth FPCA estimates a low-rank representation of $X \in \mathbb{R}^{n \times m}$ by solving

$$\min_{u,v} \|X - uv^\top\|_F^2 + \alpha\, v^\top \Omega v,$$

where $u \in \mathbb{R}^n$ denotes the scores, $v \in \mathbb{R}^m$ the loading function, Ω the roughness penalty matrix, and α the smoothing parameter. Define $S(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{m \times m}$.

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Algorithm: Penalized Power Iteration

1. Initialize v.
2. Repeat until convergence:
   1. $u \leftarrow Xv$
   2. $v \leftarrow S(\alpha) X^\top u$
   3. $v \leftarrow \dfrac{v}{\|v\|}$
3. Update $X \leftarrow X - \sigma\, uv^\top$ and proceed to next component.

Sparse Functional PCA

- *Standard FPCA may be difficult to interpret and unstable in high-dimensional settings*, since dense components can include noisy and uninformative regions, especially when $m \gg n$.

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Algorithm: sFPCA-rSVD

1. Initialization: Compute the best rank-one approximation of X using SVD
2. Iterate until convergence:
   1. Update Left Singular Vector: $u \leftarrow Xv$
   2. Update Right Singular Vector: $v \leftarrow h_\gamma X^\top u$
   3. Normalize Right Singular Vector: $v \leftarrow \frac{v}{\|v\|}$

Cross-Validation for Sparsity Selection

- **Sparsity parameter**: Tuning parameter γ controlling number of non-zero loadings in v ($0 =$ dense, $m =$ full sparsity).

Algorithm: K-fold CV Tuning Parameter Selection - Degree of sparsity

1. Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, ..., X^K$.
2. For each sparse tuning parameter $j \in \{0, 1, ..., m\}$ (level of sparsity), do the following:
   1. For $k = 1, ..., K$, let X^{-k} be the data matrix X leaving out X^k. Apply Algorithm sFPCA-rSVD on X^{-k} and derive the FPC scores $u^{-k}(j)$. Then project X^k onto $u^{-k}(j)$ to obtain $v^k(j)$.
   2. Calculate the K-fold CV scores defined as: (N is the number of grid points in X^k)
$$CV_j = \sum_{k=1}^{K} \frac{\|X^k - u^{-k}(j)v^k(j)\|^2}{N}$$
3. Select the degree of sparsity as $j_0 = \arg\min\{CV(j)\}$.

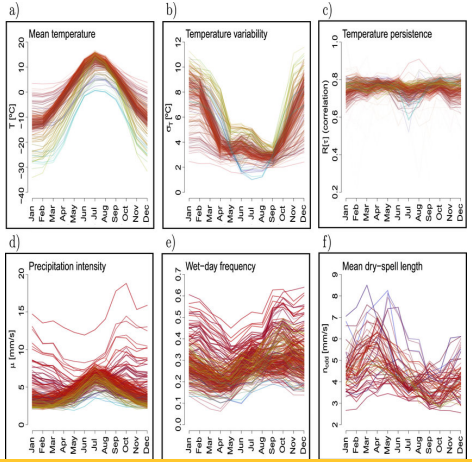
Overview of Existing Approaches

- **Smooth FPCA:**
  - Pros: Produces smooth eigenfunctions.
  - Algorithm: Penalized power iteration.
  - Tuning: α (smoothness) via GCV.
- **Sparse FPCA:**
  - Pros: Feature selection $\rightarrow$ interpretable.
  - Algorithm: sFPCA-rSVD algorithm.
  - Tuning: γ (sparsity) via CV.
- **Combined Approaches:** Smooth + Sparse together.

$$\min_{u,v} \|X - uv^\top\|_F^2 + \alpha v^\top \Omega v + p_\gamma(v)$$

Regularized MFPCA

- **Context**
  - Univariate FDA → **Multivariate FDA** → joint modes of variation across functions



Regularized MFPCA

- **Context**
  - Univariate FDA $\rightarrow$ **Multivariate FDA** $\rightarrow$ joint modes of variation across functions
- **Challenges**
  - Discretization & irregular grids $\rightarrow$ noise, missing data
  - High dimensionality and limited sample size $\rightarrow$ unstable eigenfunctions
  - Cross-function correlation $\rightarrow$ requires enforcing smoothness both within and across functions
- **Proposed Solution: Penalized SVD**
  - **Smoothness penalties**: roughness on derivatives
  - **Sparsity penalties**: Soft, hard, or SCAD
  - **Block-diagonal roughness matrix** for cross-function structure
- **Impact**
  - Produces **smooth, sparse, interpretable** joint modes
  - More stable & applicable to high-dimensional multivariate FDA

Methodology: Multivariate Functional Data Framework

- A multivariate functional dataset is formed by **concatenating** p **functional data matrices**.
  - Each variable: $X_i \in \mathbb{R}^{n \times m_i}$ where n: number of observations and m_i: grid points
- **Rank-one approximation** (per variable):

$$X_i \approx u_i v_i^\top, \quad u_i \in \mathbb{R}^n, \; v_i \in \mathbb{R}^{m_i}$$

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- **Full data matrix**: $\mathbf{X} = \begin{bmatrix} X_1 & X_2 & \cdots & X_p \end{bmatrix}$

$$\mathbf{X} = \begin{bmatrix} x_{11}(t_{11}) & \cdots & x_{11}(t_{1m_1}) & \cdots & x_{1p}(t_{p1}) & \cdots & x_{1p}(t_{pm_p}) \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1}(t_{11}) & \cdots & x_{n1}(t_{1m_1}) & \cdots & x_{np}(t_{p1}) & \cdots & x_{np}(t_{pm_p}) \end{bmatrix}$$

Penalized Smooth MFPCA

- Standard FPCA loadings may be noisy; smoothness penalties (via block-diagonal Ω_i) improve structure and interpretability.
- Let $\boldsymbol{X} \in \mathbb{R}^{n \times M}$ denote multivariate functional data, where $M = \sum_{i=1}^{p} m_i$.
  Its best rank-one approximation is $\boldsymbol{X} \approx uv^\top$,
  with $u \in \mathbb{R}^n$ (score vector) and $v = [v_1, v_2, ..., v_p]^\top \in \mathbb{R}^M$ (loading vector).
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- The block-diagonal penalty matrix is $\boldsymbol{\Omega} = \mathrm{diag}(\Omega_1, \Omega_2, \ldots, \Omega_p)$, where each $\Omega_i \in \mathbb{R}^{m_i \times m_i}$ is a univariate roughness penalty matrix.

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- The penalized reconstruction error is

$$\min_{u,v} \|\boldsymbol{X} - uv^\top\|_F^2 + \boldsymbol{\alpha}^\top (v^\top \boldsymbol{\Omega} v),$$

  where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_p)^\top$ controls smoothness.

MFPCA Power Algorithm

Algorithm: Regularized Power Iteration for Smooth MFPCA

1. Initialize v.
2. Repeat until convergence:
   1. $u \leftarrow \mathbf{X}v$
   2. $v \leftarrow \boldsymbol{S}(\boldsymbol{\alpha})\mathbf{X}^\top u$
   3. $v \leftarrow v/\|v\|$
3. Update $\mathbf{X} \leftarrow \mathbf{X} - \sigma uv^\top$ to extract the next PC.

- The smoothing operator is $\boldsymbol{S}(\boldsymbol{\alpha}) = (I + \boldsymbol{\alpha}\boldsymbol{\Omega})^{-1} \in \mathbb{R}^{M\times M}$.

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- The smoothing operator is $\boldsymbol{S}(\boldsymbol{\alpha}) = (I + \boldsymbol{\alpha}\boldsymbol{\Omega})^{-1} \in \mathbb{R}^{M\times M}$.
- The smoothing parameter $\boldsymbol{\alpha}$ is selected via generalized cross-validation (GCV), defined as

$$\mathrm{GCV}(\boldsymbol{\alpha}) = \frac{1}{M}\frac{\|(I - \boldsymbol{S}(\boldsymbol{\alpha}))(\mathbf{X}^T u)\|^2}{\left(1 - \frac{1}{M}\mathrm{tr}\{\boldsymbol{S}(\boldsymbol{\alpha})\}\right)^2}. \tag{2}$$

Penalized Sparse Multivariate FPCA

- **Goal:** Extend sparse FPCA to multivariate functional data, imposing **sparsity** (select important regions) and **smoothness** (reduce noise).
- **Sparsity penalties:** Soft, hard, or SCAD thresholding Shen and Huang [2008], Zhenhua Lin and Wang [2017], Nie and Cao [2020].

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- Sparsity parameters: $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_p)$, where γ_i ranges from 0 (no sparsity) to m_i for each variable.

Algorithm: Regularized Power Iteration for Sparse MFPCA

1. Initialization: Compute rank-one SVD of $\boldsymbol{X}$, $\boldsymbol{X} \approx duv^\top$, and set $u \leftarrow du$.
2. Iterate until convergence:
   1. Update left singular vector: $u \leftarrow \boldsymbol{X}v$
   2. Update right singular vector: $v \leftarrow \boldsymbol{h}_{\boldsymbol{\gamma}} \boldsymbol{X}^\top u$
   3. Normalize right singular vector: $v \leftarrow \dfrac{v}{\|v\|}$

Smooth and Sparse Multivariate FPCA

- The combined implementation of smoothness and sparsity on the loading vector v in multivariate functional data is achieved by the following algorithm:

Algorithm: Regularized Power Iteration for Smooth MFPCA

1. Initialize unit vectors u and v using SVD of $\boldsymbol{X}$ (best rank-one approximation of $\boldsymbol{X}$)
2. Repeat till convergence
   1. $u \leftarrow \boldsymbol{X} v$
   2. $v \leftarrow \boldsymbol{S}(\alpha)\boldsymbol{h}(\gamma_{\boldsymbol{v}})\boldsymbol{X}^\top u$
   3. $v \leftarrow \frac{v}{\|v\|}$
3. Update $\boldsymbol{X} = \boldsymbol{X} - \sigma u v^\top$ and proceed to find the next principal component.

- Algorithm CV Tuning for Sparsity and equation (2) are used to tune the sparsity level via K-fold CV and the smoothing parameter via GCV, respectively.

Simulation: Estimation Performance

- **Data-generating process:** Two functional variables:

$$X_{ij}^{(1)} = u_{i1} v_{11}(t_j) + u_{i2} v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1} v_{21}(t_j) + u_{i2} v_{22}(t_j) + \epsilon_{ij}^{(2)},$$

  - where $u_{i1} \sim N(0, \sigma_1^2)$, $u_{i2} \sim N(0, \sigma_2^2)$, $\epsilon_{ij}^{(k)} \sim N(0, \sigma^2)$, and $n = m = 101$, $t_j \in [-1, 1]$

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- **True functional PCs:**
  - Variable 1: $v_{11}(t) = \frac{t + \sin(\pi t)}{s_1}$, $\quad v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$
  - Variable 2:

$$v_{21}(t) = \begin{cases} \dfrac{\sin(3\pi t)}{s_3}, & t \in (-\frac{1}{3}, \frac{1}{3}), \\ 0, & \text{otherwise}, \end{cases} \qquad v_{22}(t) = \begin{cases} \dfrac{\sin(2\pi t)}{s_4}, & t \leq -\frac{1}{3}, \\ \dfrac{\sin(\pi t)}{s_4}, & t \geq \frac{1}{3}, \\ 0, & \text{otherwise}. \end{cases}$$

Here, s_1, s_2, s_3, s_4 are normalizing constants ensuring unit L^2 norm.

Simulation: Estimation Performance

Scenarios tested:

1. Unpenalized Multivariate SVD (baseline)
2. Smoothed Multivariate SVD (smoothness penalty)
3. Sparse Multivariate SVD (sparsity penalty)
4. Sparse + Smoothed Multivariate SVD (combined regularization)

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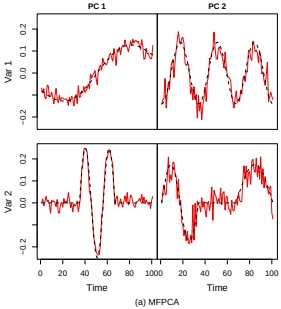

(a) MFPCA

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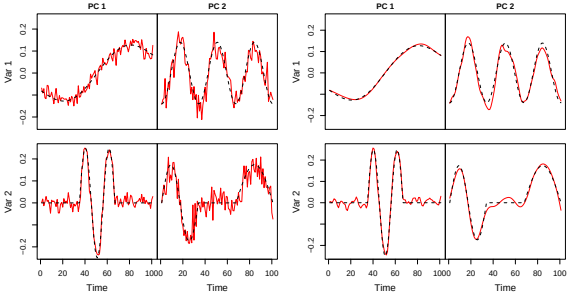

(a) MFPCA (b) Smoothed MFPCA

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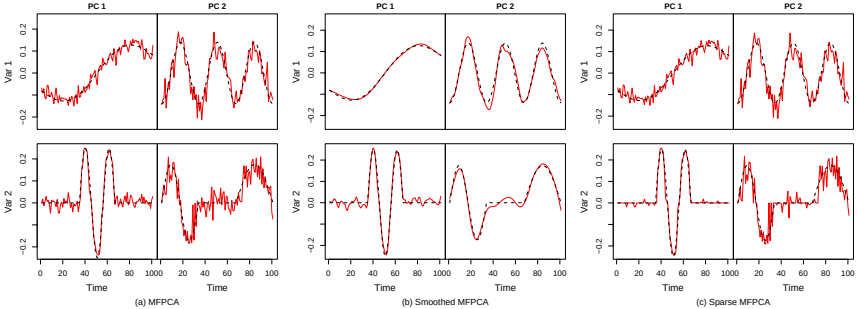

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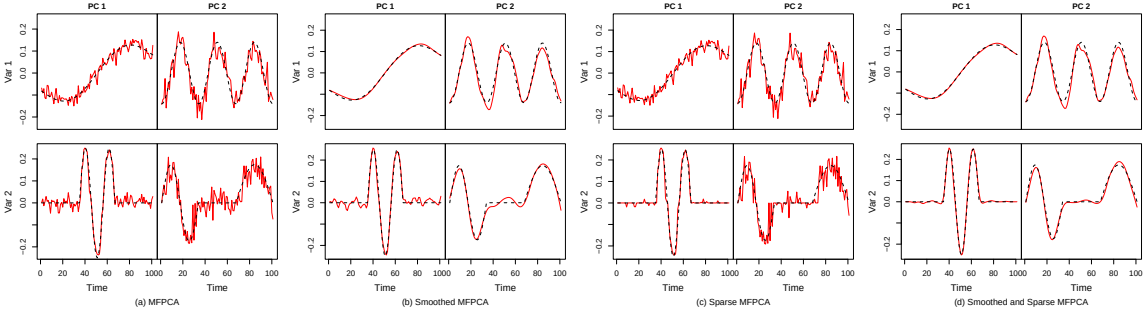


(a) MFPCA (b) Smoothed MFPCA (c) Sparse MFPCA (d) Smoothed and Sparse MFPCA

Simulation: Estimation Performance

Accuracy measures:

1. Variable-wise MSE:

$$\mathrm{MSE}_{k\ell} = \frac{1}{m}\sum_{j=1}^{m}(\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

2. Replication-averaged MSE:

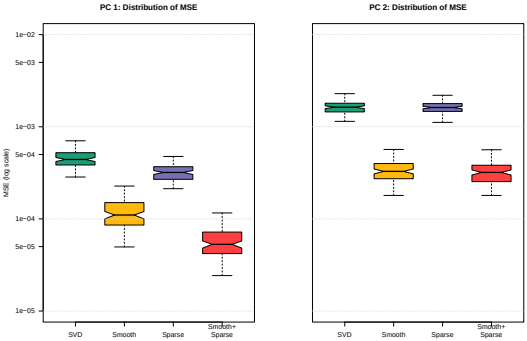
$$\overline{\mathrm{MSE}}_{k\ell} = \frac{1}{R}\sum_{r=1}^{R}\mathrm{MSE}_{k\ell}^{(r)}$$

3. Multivariate MSE:

$$\mathrm{MSE}_{\ell}^{(\mathrm{multi})} = \frac{1}{m}\sum_{j=1}^{m}\sum_{k=1}^{2}(\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

Simulation: Estimation Performance

- **Performance across four methods (SVD, Smooth, Sparse, Smooth+Sparse):**
  - Smoothness and/or sparsity **reduce MSE** compared to unregularized SVD.
  - **Smooth+Sparse** yields lowest error and most stable estimates.
  - Smooth estimator performs consistently well; sparsity alone less effective (esp. for PC2).
  - Joint regularization achieves best **bias–variance tradeoff**.



PC1: Quartiles and Mean log10(MSE)

Method	Q1	Median	Mean	Q3
SVD	-3.41	-3.35	-3.34	-3.28
Smooth	-4.07	-3.96	-3.92	-3.82
Sparse	-3.57	-3.50	-3.49	-3.43
Smooth+Sparse	**-4.38**	**-4.28**	**-4.22**	**-4.14**

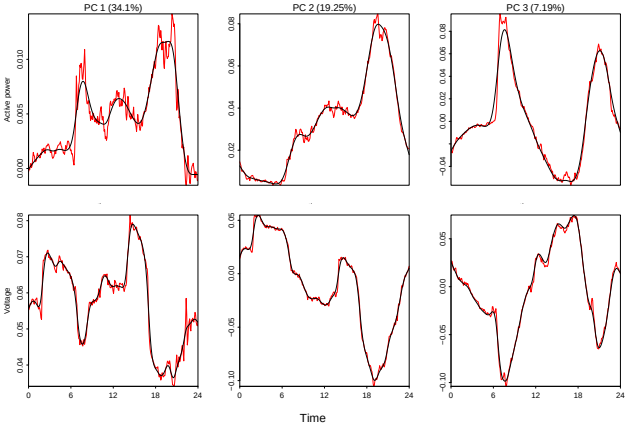
PC2: Quartiles and Mean log10(MSE)

Method	Q1	Median	Mean	Q3
SVD	-2.84	-2.79	-2.79	-2.75
Smooth	-3.56	-3.48	-3.47	-3.40
Sparse	-2.83	-2.79	-2.79	-2.75
Smooth+Sparse	**-3.59**	**-3.50**	**-3.49**	**-3.42**

Application: Household Power Consumption

- **Dataset:** Bivariate functional data including active power and voltage consumption [Hebrail and Berard, 2012] for one household between December 2006 and November 2010.

- Regularization reduces noise while preserving the dominant daily consumption patterns, enhancing interpretability without losing key structure.



First 3 PCs: MFPCA (red) vs ReMFPCA (black)

Two-way Regularized MFPCA

- **Two-way functional data:** Two-way functional data consist of a data matrix whose row and column domains are both structured
- **Limitation of standard FPCA [Ramsay and Silverman, 2005]:**
  - Focuses on one domain (often time).
  - Penalties applied only to loadings $\rightarrow$ ignores structure in second domain.
  - Results may be rough or overly dense along the unpenalized axis.

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  - Penalties applied only to loadings $\rightarrow$ ignores structure in second domain.
  - Results may be rough or overly dense along the unpenalized axis.
- **Two-way FPCA [Jianhua Z. Huang and Buja, 2009]:**
  - Introduced **smoothness penalties** on both scores and loadings.
  - Produces coherent, interpretable *component surfaces* instead of jagged approximations.
- **Our contribution:**
  - Extend to **two-way multivariate functional data** (multiple functional variables).
  - Combine **smoothness + sparsity penalties** in both directions.
  - Result: Low-rank, interpretable, noise-robust PCs for high-dimensional applications.

Two-way Smoothed MFPCA: Setup & Penalty

- Two-way multivariate functional data: $\boldsymbol{X} \in \mathbb{R}^{n \times M}, \quad M = \sum_{i=1}^{p} m_i$.
- Roughness matrices: $\boldsymbol{\Omega}_u \in \mathbb{R}^{n \times n}$, $\boldsymbol{\Omega}_v \in \mathbb{R}^{M \times M}$ (symmetric, non-negative definite).
- Smoothers: $\boldsymbol{S}_u(\alpha_u) = (I + \alpha_u \boldsymbol{\Omega}_u)^{-1}, \qquad \boldsymbol{S}_v(\boldsymbol{\alpha}_v) = (I + \boldsymbol{\alpha}_v \boldsymbol{\Omega}_v)^{-1}$.

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- Penalized rank-one reconstruction:

$$\min_{u,v} \|\boldsymbol{X} - uv^\top\|_F^2 + \mathcal{P}(u, v)$$

- Penalty [Jianhua Z. Huang and Buja, 2009]:
  $\mathcal{P}(u, v; \alpha_u, \boldsymbol{\alpha}_v) = u^\top(\alpha_u \boldsymbol{\Omega}_u)u\,\|v\|^2 + \|u\|^2\, v^\top(\boldsymbol{\alpha}_v \boldsymbol{\Omega}_v)v + u^\top(\alpha_u \boldsymbol{\Omega}_u)u\; v^\top(\boldsymbol{\alpha}_v \boldsymbol{\Omega}_v)v$.
- Multivariate v: $\boldsymbol{\Omega}_v = \mathrm{diag}(\Omega_1, \ldots, \Omega_p)$.

Two-way Smoothed MFPCA: Conditional GCV

- Minimizers:

$$u = \frac{\boldsymbol{S}_u(\alpha_u)\boldsymbol{X}v}{v^\top(I+\boldsymbol{\alpha}_v\boldsymbol{\Omega}_v)\,v} = \frac{\boldsymbol{S}_u(\alpha_u)}{1+\boldsymbol{\alpha}_v\boldsymbol{R}_v(v)}\,\frac{\boldsymbol{X}v}{\|v\|^2}, \qquad v = \frac{\boldsymbol{S}_v(\boldsymbol{\alpha}_v)\boldsymbol{X}^\top u}{u^\top(I+\alpha_u\boldsymbol{\Omega}_u)\,u} = \frac{\boldsymbol{S}_v(\boldsymbol{\alpha}_v)}{1+\alpha_u\boldsymbol{R}_u(u)}\,\frac{\boldsymbol{X}^\top u}{\|u\|^2}.$$

- Rayleigh quotients: $\boldsymbol{R}_u(u) = \frac{u^\top\boldsymbol{\Omega}_u u}{\|u\|^2}, \qquad \boldsymbol{R}_v(v) = \frac{v^\top\boldsymbol{\Omega}_v v}{\|v\|^2}.$

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- Rayleigh quotients: $\boldsymbol{R}_u(u) = \frac{u^\top\boldsymbol{\Omega}_u u}{\|u\|^2}, \qquad \boldsymbol{R}_v(v) = \frac{v^\top\boldsymbol{\Omega}_v v}{\|v\|^2}.$
- Conditional GCV criteria [Jianhua Z. Huang and Buja, 2009]:

$$\mathrm{GCV}_u(\alpha_u;\boldsymbol{\alpha}_v) = \frac{\frac{1}{n}\left\|\left(I-\frac{\boldsymbol{S}_u(\alpha_u)}{1+\boldsymbol{\alpha}_v\boldsymbol{R}_v(v)}\right)\frac{\boldsymbol{X}v}{\|v\|^2}\right\|^2}{\left(1-\frac{1}{n}\operatorname{tr}\left(\frac{\boldsymbol{S}_u(\alpha_u)}{1+\boldsymbol{\alpha}_v\boldsymbol{R}_v(v)}\right)\right)^2}, \qquad \mathrm{GCV}_v(\boldsymbol{\alpha}_v;\alpha_u) = \frac{\frac{1}{m}\left\|\left(I-\frac{\boldsymbol{S}_v(\boldsymbol{\alpha}_v)}{1+\alpha_u\boldsymbol{R}_u(u)}\right)\frac{\boldsymbol{X}^\top u}{\|u\|^2}\right\|^2}{\left(1-\frac{1}{m}\operatorname{tr}\left(\frac{\boldsymbol{S}_v(\boldsymbol{\alpha}_v)}{1+\alpha_u\boldsymbol{R}_u(u)}\right)\right)^2}.$$

- **Optimization:** Alternate updates of u and v using GCV until convergence $\rightarrow$ two-way regularized components.

Two-way Smooth + Sparse MFPCA

- **Goal:** Extract components that are **low-rank, smooth, and sparse**.
  - Smoothness $\rightarrow$ coherent variation across subjects & functions.
  - Sparsity $\rightarrow$ highlights key observations & time regions.
- **Novelty:** First framework to combine **both** in two-way functional data.

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  - Smoothness $\rightarrow$ coherent variation across subjects & functions.
  - Sparsity $\rightarrow$ highlights key observations & time regions.
- **Novelty:** First framework to combine **both** in two-way functional data.
- Data matrix $\boldsymbol{X}$: seek u, v solving:

$$\min_{u,v} \|\boldsymbol{X} - uv^\top\|_F^2 + \sum_{j}^{J} \mathcal{P}_j^{[\theta]}(u, v)$$

- J is the number of penalty components, and θ is the vector of all tuning parameters.
- The composite penalty $\sum_{j=1}^{J} \mathcal{P}_j^{(\theta)}(u, v)$ lets us mix regularizers, e.g., smoothness with $\theta = (\alpha_u, \boldsymbol{\alpha}_v)$ and sparsity with $\theta = (\gamma_u, \boldsymbol{\gamma}_v)$ (controlling sparsity), and can include other structures as needed.

Sequential Power Algorithm

Algorithm: Two-way Smooth + Sparse MFPCA (Sequential Power)

1. Initialization: Rank-one SVD of $\boldsymbol{X}$: $\boldsymbol{X} \approx s\, u^{(0)} v^{(0)\top}$; set $u \leftarrow s\, u^{(0)}$, $v \leftarrow v^{(0)}$.
2. Repeat until convergence:
   1. $u \leftarrow \boldsymbol{S}_u^{[\alpha_u]}\, \boldsymbol{h}_u^{[\gamma_u]}(\boldsymbol{X}\, v)$
   2. $v \leftarrow \boldsymbol{S}_v^{[\boldsymbol{\alpha}_v]}\, \boldsymbol{h}_v^{[\gamma_v]}(\boldsymbol{X}^\top u)$
   3. $v \leftarrow v / \|v\|$
3. $\boldsymbol{X} \leftarrow \boldsymbol{X} - \sigma\, u\, v^\top$ to extract the next component.

- Smoothness parameters are selected with conditional GCV, while sparsity parameters are chosen via cross-validation (CV).

Selection of Regularization Parameters

- Four sets of tuning parameters:
  - α_u: smoothness of u, γ_u: sparsity of u
  - $\boldsymbol{\alpha}_v$: smoothness of v, $\boldsymbol{\gamma}_v$: sparsity of v
- **Challenge:** Ordering of tuning (smoothness vs. sparsity) affects convergence and solutions.

Selection of Regularization Parameters

- Four sets of tuning parameters:
  - α_u: smoothness of u, γ_u: sparsity of u
  - $\boldsymbol{\alpha}_v$: smoothness of v, $\boldsymbol{\gamma}_v$: sparsity of v
- **Challenge:** Ordering of tuning (smoothness vs. sparsity) affects convergence and solutions.
- **Strategy: Conditional tuning**
  1. Initialize all penalties at 0.
  2. Tune γ_u via K-fold CV.
  3. Sequentially tune $\gamma_{v,i}$ using Algorithm:Two-way Smooth + Sparse MFPCA.
  4. With sparsity fixed, tune α_u by GCV.
  5. Tune $\alpha_{v,i}$ using two-way GCV.
  6. Iterate steps 2–5 until stable.
- This alternating scheme **isolates sparsity vs. smoothness** while ensuring accuracy + interpretability.

K-Fold CV algorithm for Sparsity

K-Fold CV (Row Sparsity)

1. Split $\boldsymbol{X} \in \mathbb{R}^{n \times M}$ into K column groups $\{\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(K)}\}$.
2. For each γ_j and $k = 1, \ldots, K$:
   1. Train on $\boldsymbol{X}^{(-k)}$, estimate $u_j^{(-k)}$.
   2. Validate: $v_j^{(k)} = \boldsymbol{X}^{(k)\top} u_j^{(-k)}$.
   3. Fold error:
   $$\mathrm{Err}_j^{(k)} = \frac{1}{\tilde{M}} \|\boldsymbol{X}^{(k)} - u_j^{(-k)} (v_j^{(k)})^\top\|_F^2.$$
3. CV score: $\widehat{\mathrm{CV}}_j = \frac{1}{K} \sum_k \mathrm{Err}_j^{(k)}$.
4. Select $j_0 = \arg\min_j \widehat{\mathrm{CV}}_j$.

K-Fold CV + 1-SE Rule

1. Use same folds to collect $\mathrm{Err}_j^{(k)}$.
2. Compute mean $\widehat{\mathrm{CV}}_j$ and SE $\widehat{\mathrm{SE}}_j$:
$$\widehat{\mathrm{SE}}_j = \sqrt{\frac{1}{K(K-1)} \sum_k (\mathrm{Err}_j^{(k)} - \widehat{\mathrm{CV}}_j)^2}.$$
3. Let $j^\star = \arg\min_j \widehat{\mathrm{CV}}_j$.
4. Choose sparsest j_0 with $\widehat{\mathrm{CV}}_j \leq \widehat{\mathrm{CV}}_{j^\star} + \widehat{\mathrm{SE}}_{j^\star}$.

K-Fold CV algorithm for Sparsity

K-Fold CV (Column Sparsity)

1. Split $\boldsymbol{X} \in \mathbb{R}^{n \times M}$ into K row groups $\{\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(K)}\}$.
2. For each γ_j and $k = 1, \ldots, K$:
   1. Train on $\boldsymbol{X}^{(-k)}$, estimate $v_j^{(-k)}$.
   2. Validate: $u_j^{(k)} = \boldsymbol{X}^{(k)} v_j^{(-k)}$.
   3. Fold error:
   $$\mathrm{Err}_j^{(k)} = \frac{1}{n} \|\boldsymbol{X}^{(k)} - u_j^{(k)} (v_j^{(-k)})^\top\|_F^2.$$
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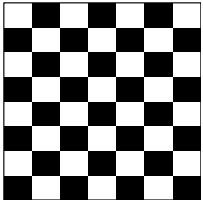
Chessboard

True Chessboard



Chessboard

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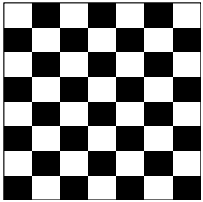
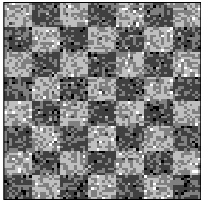


Noisy Chessboard



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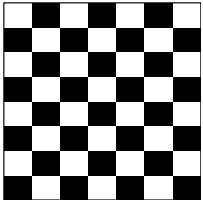
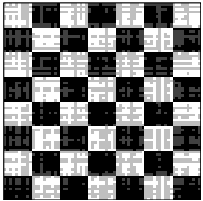
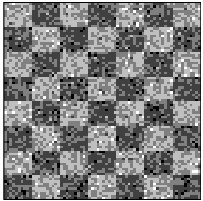


PCA with SVD

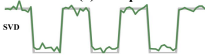


First PC Scores (u) Comparison

Noisy Chessboard



First PC (v) Comparison



Chessboard

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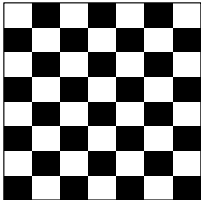

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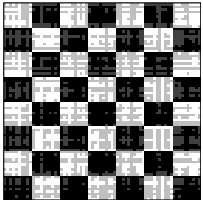
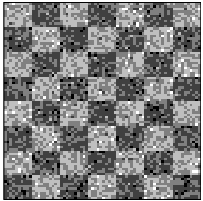
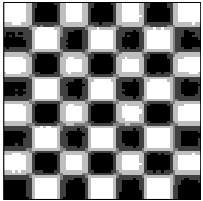


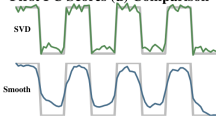
Noisy Chessboard



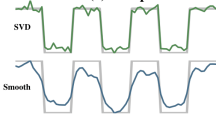
PCA with Smoothness Penalty



First PC Scores (u) Comparison

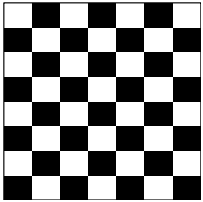


First PC (v) Comparison

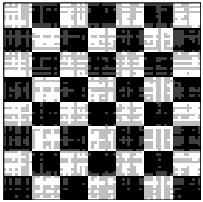


Chessboard

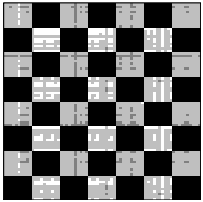
True Chessboard



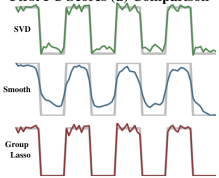
PCA with SVD



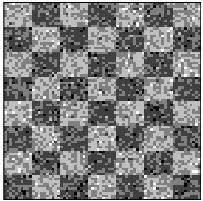
PCA with Group Lasso Penalty



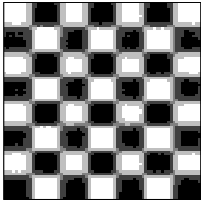
First PC Scores (u) Comparison



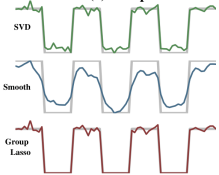
Noisy Chessboard



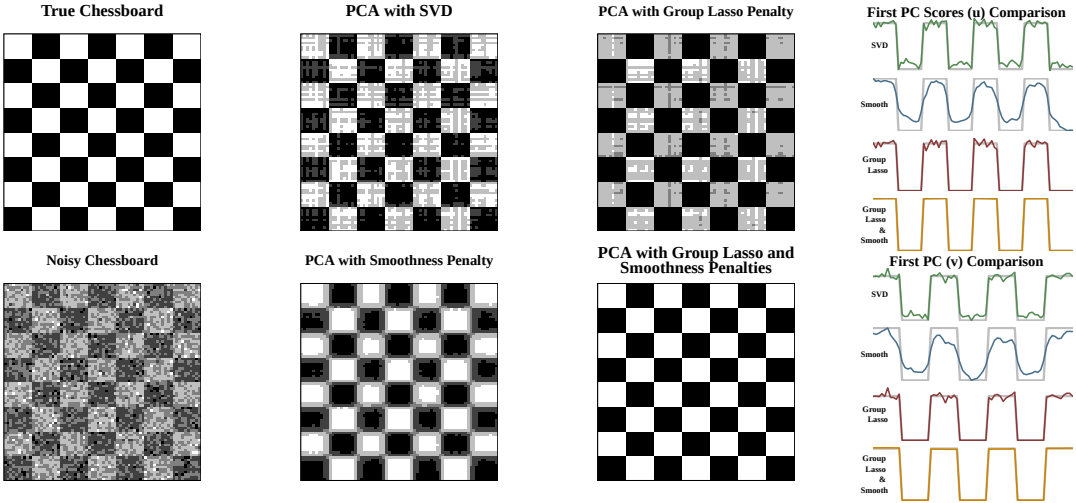
PCA with Smoothness Penalty



First PC (v) Comparison



Chessboard


True Chessboard
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SVD
Smooth
Group Lasso
Group Lasso & Smooth
Noisy Chessboard
PCA with Smoothness Penalty
PCA with Group Lasso and Smoothness Penalties
First PC (v) Comparison
SVD
Smooth
Group Lasso
Group Lasso & Smooth

Simulation: Two-way Functional Data

- **Data-generating process:**
  $X_{ij}^{(1)} = u_{i1}v_{11}(t_j) + u_{i2}v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1}v_{21}(t_j) + u_{i2}v_{22}(t_j) + \epsilon_{ij}^{(2)},$
- **Latent scores:** generated as smooth functions

$$u_1(s) = \begin{cases} \sin(\pi s), & s > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad u_2(s) = \sin(2\pi s), \quad s \in [-1, 1].$$

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- **Functional PCs:**
  - Variable 1: $v_{11}(t) = \frac{t + \sin(\pi t)}{s_1}, \qquad v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$
  - Variable 2:

$$v_{21}(t) = \begin{cases} \dfrac{\sin(3\pi t)}{s_3}, & t \in \left(-\frac{1}{3}, \frac{1}{3}\right), \\ 0, & \text{otherwise}, \end{cases} \qquad v_{22}(t) = \begin{cases} \dfrac{\sin(2\pi t)}{s_4}, & t \leq -\frac{1}{3}, \\ \dfrac{\sin(\pi t)}{s_4}, & t \geq \frac{1}{3}, \\ 0, & \text{otherwise}. \end{cases}$$

Evaluation Metrics

- **Integrated Squared Error (ISE):**
  For replicate r and component u_1:

$$\mathrm{ISE}_r^{(u_1,\text{method})} = \frac{1}{m}\sum_{j=1}^{m}\left(u_1(t_j) - \widehat{u}_1^{(\text{method})}(t_j)\right)^2.$$

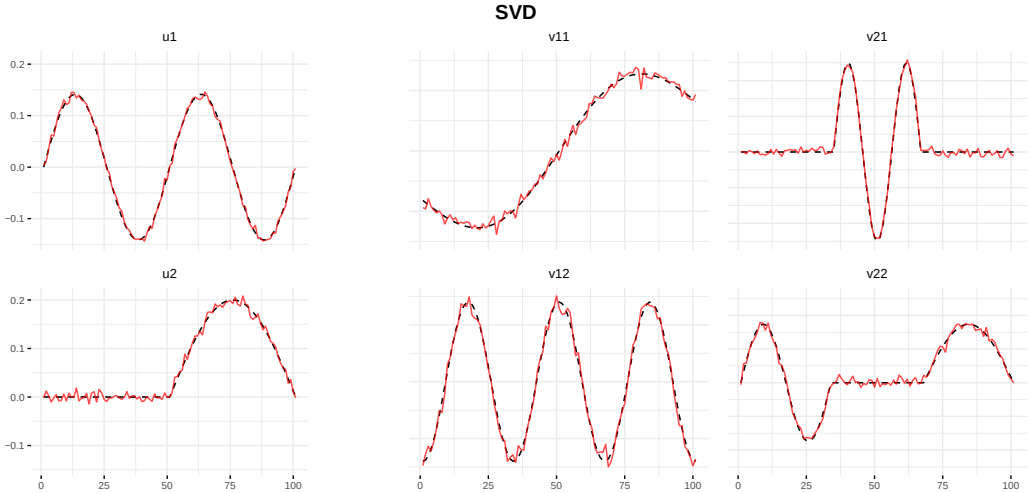
- **Relative ISE (R_ISE):** ratio vs best method

$$R_r^{(u_1,\text{method})} = \frac{\mathrm{ISE}_r^{(u_1,\text{method})}}{\mathrm{ISE}_r^{(u_1,\text{best})}}.$$

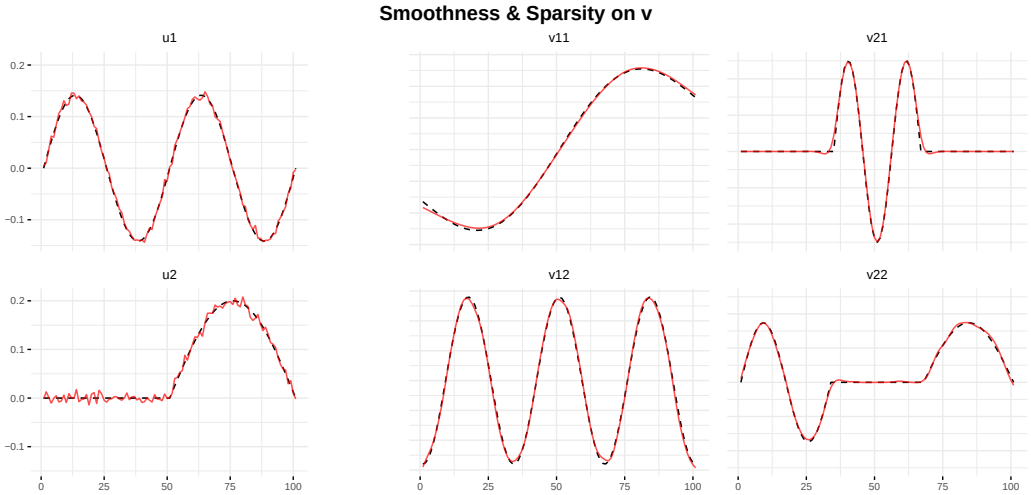
- **Monte Carlo averages:**

$$\overline{R}^{(u_1,\text{method})} = \frac{1}{N}\sum_{r=1}^{N} R_r^{(u_1,\text{method})}, \qquad \mathrm{SE}(\overline{R}) = \sqrt{\frac{1}{N(N-1)}\sum_{r=1}^{N}\left(R_r - \overline{R}\right)^2}.$$

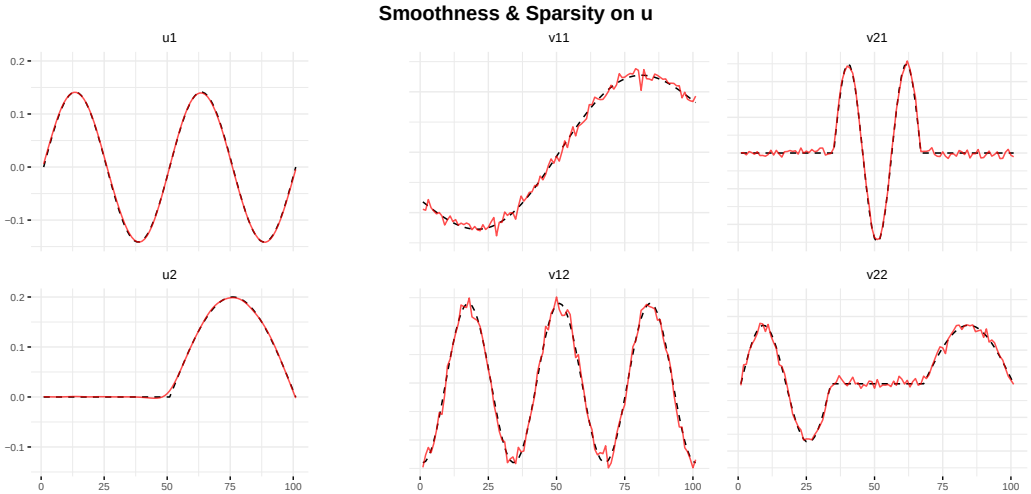
Simulation Results (SVD)



Simulation Results (Smoothness and Sparsity on v)

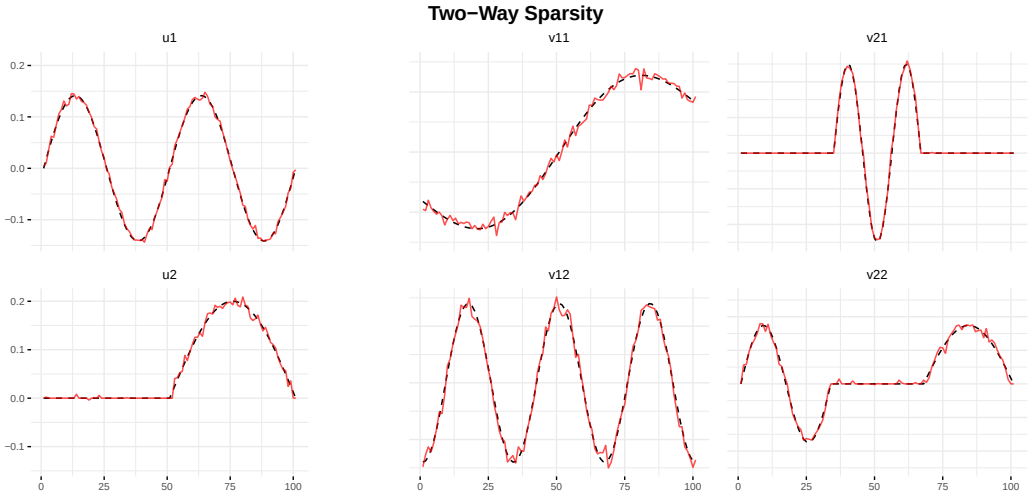


Simulation Results (Smoothness and Sparsity on u)

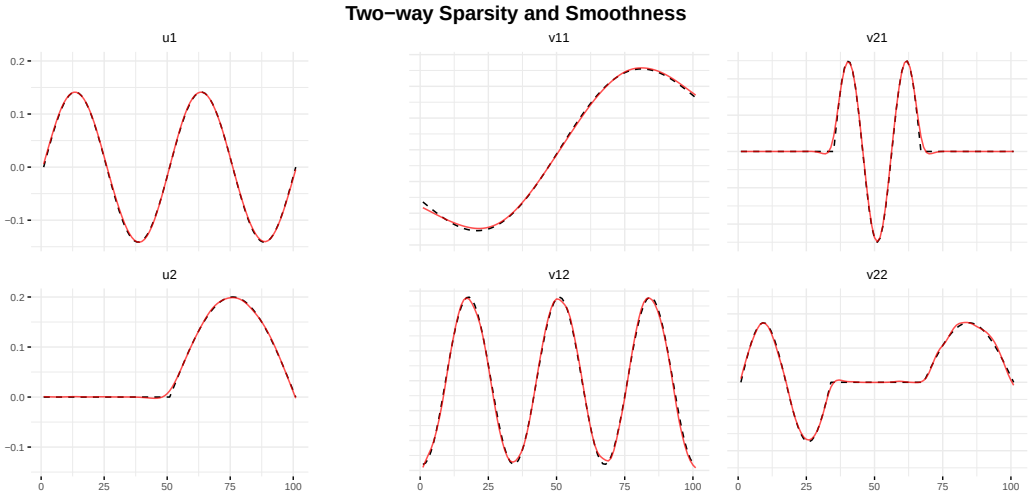


Simulation Results (Two-Way Smoothness)

Simulation Results (Two-Way Sparsity)



Simulation Results (Two-way Sparsity and Smoothness)



Simulation Results

Table 3: Mean ISE for each method and parameter

Method	u1	u2	v1	v2
SVD	0.3651	0.1587	0.00005	0.00010
Smooth+Sparse v	0.3651	0.1587	**0.00002**	**0.00002**
Smooth+Sparse u	**0.3650**	**0.1584**	0.00005	0.00010
Two-way Smoothness	**0.3650**	0.1585	**0.00002**	**0.00002**
Two-way Sparsity	0.3651	0.1586	0.00004	0.00009
Two-way Sm+Sp	**0.3650**	**0.1584**	**0.00002**	**0.00002**

Table 4: Mean Relative ISE for each method and parameter

Method	u1	u2	v1	v2
SVD	1.000	1.001	8.21	5.23
Two-way Sparsity	1.000	1.001	7.71	4.61
Smooth+Sparse v	1.000	1.001	1.05	**1.01**
Smooth+Sparse u	**1.000**	**1.000**	8.19	5.21
Two-way Smoothness	**1.000**	**1.000**	**1.00**	1.21

- Two-way Smooth+Sparse consistently yields the lowest errors across u and v.

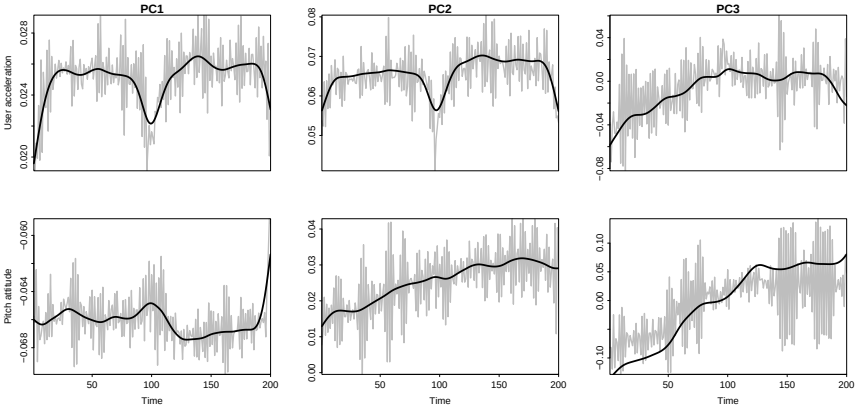
Application: Motion Sense Data

- **Dataset:** Acceleration and pitch from 24 people, 4 activities (jogging, walking, sitting, standing), about 2–3 min each.
- **Goal:** Compare SVD vs **two-way sparse + smooth ReMFPCA** on these multivariate functional signals.
- **Rescaling [Happ and Greven, 2018]:** balance variables so each contributes equally.

$$\hat{w}_j = \left(\frac{1}{m} \sum_{i=1}^{m} \widehat{\mathrm{Var}}(X_j(t_i)) \right)^{-1}, \qquad \tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i).$$

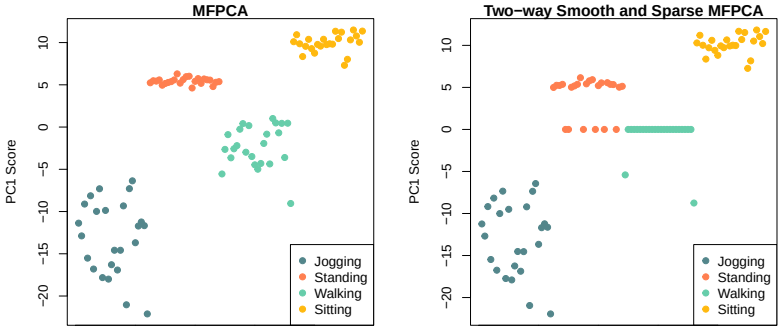
- **Penalties used:** smoothness + sparsity on loadings v; sparsity on scores u.

Results: Functional PCs (SVD vs ReMFPCA)



- **SVD (gray):** noisy, high-frequency wiggles.
- **ReMFPCA (black):** smoother, more interpretable PCs capturing dominant structure.

Results: PC Scores and Interpretation



- Sparsity on scores: **PC1 scores for walking about 0** (partially standing too).
- Interpretation: walking contributes little to PC1; removing it **improves interpretability** without hurting fit.
- Takeaway: **Two-way smooth + sparse ReMFPCA** yields cleaner PCs and activity-informative scores.

Conclusion & Future Work

- **Unified FPCA Framework**
  - Combines **smoothness** (denoise, interpretability) + **sparsity** (variable selection).
  - Extends from **univariate** → **multivariate** → **two-way functional data**.
- **Methodology**
  - Penalized SVD with roughness + ℓ_1 penalties.
  - Two-way regularization: smoothness & sparsity on both **scores (u)** and **loadings (v)**.
  - Efficient parameter tuning: **conditional GCV** & **K-fold CV** (with 1-SE rule).
- **Results**
  - Simulations & applications (mortality, call-center, image data).
  - Outperforms one-way or single-penalty methods.
  - Produces **low-rank, denoised, interpretable components**.

Accessible Implementation: R Package & Future Work

- Implemented in **R package `ReMPCA` (GitHub)**
  - Univariate & multivariate FPCA with penalties.
  - Two-way MFPCA for matrix-valued functions.
  - Automated tuning (CV, GCV, 1-SE rule).
  - Diagnostic tools: variance explained, visualization, heatmaps.
  - Early support for **hybrid data (scalar + functional + image)**.
- **Hybrid Data Extensions**
  - Image–Functional Hybrid PCA $\rightarrow$ simultaneous dimension reduction.
  - Scalar–Functional Integration $\rightarrow$ joint low-dim space.
  - Nonlinear Extensions $\rightarrow$ kernel FPCA, neural nets.
- **Applications:**
  - Neuroimaging
  - Personalized medicine
  - Environmental monitoring

  Takeaway: Smooth + sparse + two-way FPCA offers a **theoretical foundation, practical algorithms, and open software** to enable next-generation functional data analysis.

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Thank you!

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Appendix

Variance Explained: Classical vs Regularized FPCA

- **Classical FPCA:** Loadings v_j orthonormal; scores $u_j = Xv_j$ uncorrelated. Variance explained by first J PCs:

$$\sum_{j=1}^{J} \|u_j\|^2 = \mathrm{trace}(V_J^\top X^\top X\, V_J), \qquad V_J = [v_1, \ldots, v_J].$$

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- **Issue under regularization:** smoothness/sparsity break orthogonality $\rightarrow$ scores become correlated $\rightarrow$ naive sum $\sum \|u_j\|^2$ **double-counts** variance (cf. Huang et al., 2008).

Appendix

Subspace-Projection Definition of Explained Variance

- Normalize loadings and stack:

$$V_J = [v_1, \ldots, v_J], \qquad v_j \leftarrow \frac{v_j}{\|v_j\|}.$$

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Subspace-Projection Definition of Explained Variance

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- Orthogonal projector matrix onto span $(v_1, \ldots, v_J)$:

$$H_J = V_J \left(V_J^\top V_J\right)^{-1} V_J^\top,$$

where H_J is a a symmetric idempotent matrix. $(H_J^2 = H_J,\ H_J^\top = H_J)$

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- Projected data and explained variance:

$$X_J = XH_J, \qquad \mathrm{V}_{\mathrm{tot}} = \mathrm{tr}(X^\top X), \qquad \mathcal{V}_J = \|X_J\|_F^2 = \mathrm{tr}(H_J X^\top X H_J).$$

Appendix

PVE, Incremental PVE, and Properties

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$$\Delta\mathcal{V}_j = \mathcal{V}_j - \mathcal{V}_{j-1}, \qquad \mathcal{V}_0 = 0.$$

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$$\mathrm{PVE}(J) = \frac{\mathcal{V}_J}{\mathsf{V}_{\mathsf{tot}}}, \qquad \mathrm{pve}_j = \frac{\Delta\mathcal{V}_j}{\mathsf{V}_{\mathsf{tot}}} = \mathrm{PVE}(j) - \mathrm{PVE}(j-1).$$

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- **Key properties:**
  - No double-counting (works with correlated scores).
  - Reduces to classical PCA when $V_J^\top V_J = I_J$.
  - Monotone in J ($\mathcal{V}_J$ increases).
  - $\Delta\mathcal{V}_j$ = unique variance added by component j.